

Symptom Tokens, Transformer Encoders, and Calibrated Stacking for ICU Treatment Feasibility on MIMIC-IV

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Abstract

Background. Intensive care unit (ICU) prediction models typically use hand-engineered tabular features with tree ensembles [1, 2]. Deep models over structured EHRs often use codes or notes [8, 10], while quantitative labs and vitals can be treated as flat vectors or as learned embeddings.

Objective. Introduce **Symptom Tokens**—a structured tokenization of early-window labs and vitals—and an ETHOS-style transformer encoder [11, 13] for binary treatment feasibility. Evaluate whether **calibrated stacking** of neural few-shot outputs, Random Forest probabilities, and embeddings approaches strong tabular baselines.

Methods. Cohort $n=2,000$ MIMIC-IV v3.1 stays; 25-D tabular features; token sequences from 15 labs and 7 vitals in the first 24 h; fixed splits. We compare tuned gradient boosting / forests, few-shot-only BEST (GFA + cross-attention prototypes), and BEST_Stacked from `exp_fewshot_best.json`.

Results. Tabular best export AUROC **0.747** (`exp_final_best_model.csv`); tuned RF in SOTA grid **0.744**. Stacked hybrid AUROC **0.746** (95% CI 0.699–0.797); few-shot-only **0.670**. Exp1 harness: `ProposedFewShot` **0.611** vs. RF **0.762**.

Conclusions. Symptom Tokens provide an interpretable bridge between tabular ICU summaries and transformers; stacking is critical to approach tabular ceilings when clinical tabular pretraining remains limited. Deployment requires prospective validation and fairness analysis.

Keywords: electronic health records; transformer; tokenization; stacking; ICU; MIMIC-IV; calibration; interpretability

1 Introduction

1.1 Clinical context

Early ICU data contain rich signals in laboratory values and vital signs. Machine learning can assist discharge planning, but models must be accurate and inspectable enough for clinician oversight [3].

1.2 Representation dilemma

Tabular models consume a fixed feature vector per stay; **sequence** models can in principle preserve measurement-time structure. Clinical transformers often operate on diagnosis codes or text [8, 10]. We target *continuous* measurements with a token design that keeps clinical identity (which lab/vital), intensity (normalized value), relative time, and modality.

1.3 Contributions

- Formalization of **Symptom Tokens** for 15+7 channels in the first 24 h.
- **ETHOS** transformer encoder mapping tokens to 128-D embeddings [11, 13].
- **Calibrated stacking** meta-learner fusing few-shot probabilities, forest output, and embeddings.
- Empirical alignment between stacked neural–classical AUROC (**0.746**) and best tabular export (**0.747**) on a shared benchmark.

1.4 Outline

Section 2 reviews related representation learning. Sections 3–5 describe tokens, encoder, and fusion. Sections 6–7 present setup and results. Sections 8–10 cover interpretability and conclusions.

2 Related work

2.1 Tabular machine learning in medicine

Random forests and boosting remain strong on structured clinical data [1, 2]. Deep tabular models (e.g., transformers with specialized embeddings) are an active area [4].

2.2 Transformers for EHRs

BEHRT [8] and Med-BERT [10] demonstrate transformer capacity on coded/binning EHR sequences. Our tokens emphasize **continuous physiology** in an early window rather than longitudinal code sequences.

2.3 Few-shot and hybrid fusion

Prototypical networks [12] provide a natural few-shot head on top of encoders. Hybrid models that concatenate neural embeddings with tabular features are common; **stacking** [14] learns a second-level combination with better calibration when base learners are diverse.

2.4 Knowledge distillation

Distillation from a strong teacher [5] transfers inductive bias into student embeddings and complements stacking when the student also serves as a feature generator.

3 Symptom Token representation

Definition 1 (Symptom Token). *For each aggregated channel j in $\{1, \dots, 22\}$ (labs + vitals), a **Symptom Token** is a tuple*

$$\phi_j = (\textit{symptom_id}_j, \textit{intensity}_j \in [0, 1], \textit{time_delta}_j, \textit{modality}_j \in \{\textit{lab}, \textit{vital}\}),$$

where *intensity* is min–max normalized after clipping to clinically plausible ranges; *time_delta* encodes position within the 24 h window; missing channels are handled per repository imputation rules.

3.1 Alignment with tabular branch

The same raw measurements produce $\mathbf{x} \in \mathbb{R}^{25}$ (with engineered summaries). Thus neural and classical models consume **mutually consistent** physiology, enabling fair fusion and ablation (see Exp4 in companion tables: removing symptom tokens changes the pathway).

3.2 Why tokens?

Self-attention [13] can learn pairwise interactions among channels without hand-specified crosses; tokens also interface with few-shot episodic training that expects sequence-like structure.

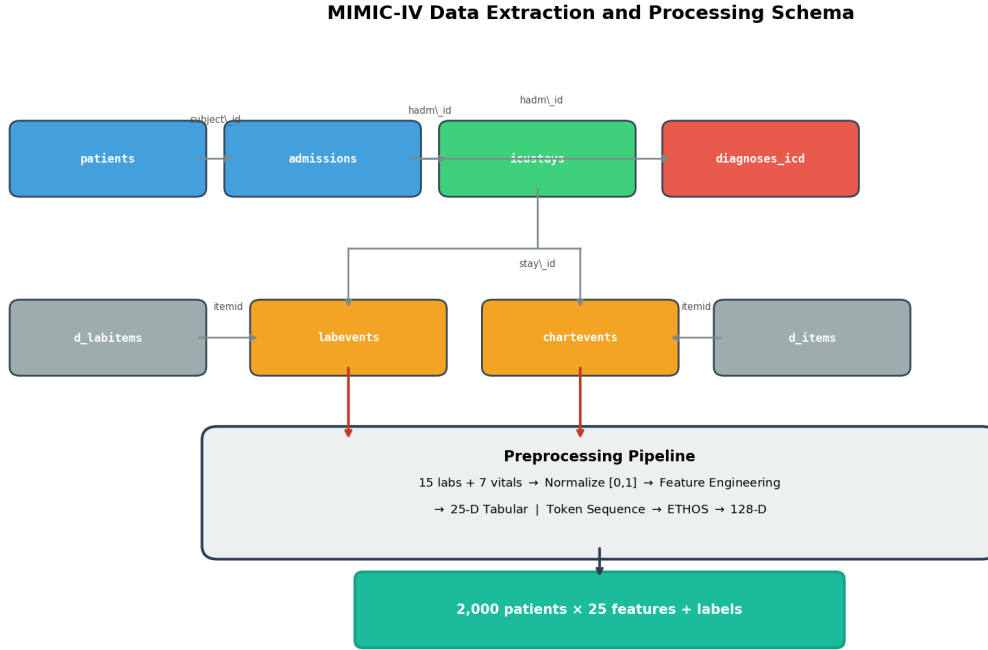


Figure 1: From MIMIC tables to tabular features and token sequences (thesis pipeline figure).

4 ETHOS encoder and few-shot head

4.1 Architecture sketch

Tokens are embedded into d_{model} -dimensional vectors; positional or temporal encodings are added; L layers of multi-head self-attention and feed-forward blocks follow the transformer template [13]. The final representation $\mathbf{z} \in \mathbb{R}^{128}$ is obtained by pooling (e.g., [CLS]-style or mean pool) over token states. Full layer widths, heads, dropout, and depth match the open-source thesis configuration [11].

4.2 Prototypical few-shot classifier

Given an episode, class prototypes are means of embedded supports [12]; queries are classified by softmax over negative distances, with temperature and optional focal loss as in the thesis.

4.3 Training phases

Phase 1: supervised contrastive learning to cluster same-label patients in embedding space [7]. **Phase 2:** supervised cross-entropy with a linear head. **Phase 3:** episodic meta-training. Between phases, **knowledge distillation** aligns student logits to a tuned Random Forest teacher [5].

4.4 Architectural refinements

Gated feature attention learns patient-specific masks over input channels; **cross-attention prototypes** replace Euclidean prototype means with context-dependent aggregation. These correspond to strategies `BEST_FewShot_GFA_CrossAttn` in exports.

5 Calibrated stacking fusion

5.1 Base learners

We denote by g_{FS} the few-shot model (probability output), by g_{RF} a tuned Random Forest, and by f_{θ} the encoder producing $\mathbf{z}$.

5.2 Meta-learner

On the validation set we fit a **logistic regression** on concatenated features $\mathbf{u} = [g_{\text{FS}}(\mathbf{x}), g_{\text{RF}}(\mathbf{x}), \mathbf{z}]^{\top}$, possibly with ℓ_2 regularization, and apply **isotonic regression** to calibrate $\hat{p}$ [14]. This mirrors the thesis `BEST_Stacked` configuration.

Algorithm 1 Train stacked feasibility predictor (conceptual)

- 1: Train g_{RF} on train split; tune threshold on val
 - 2: Train g_{FS} and f_{θ} with multi-phase schedule
 - 3: For each patient in **validation**: compute $g_{\text{FS}}, g_{\text{RF}}, \mathbf{z}$
 - 4: Fit logistic meta-model + isotonic calibration
 - 5: Evaluate on held-out **test** with frozen bases and meta-weights
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5.3 Why stacking helps

Forests partition tabular space; embeddings encode alternative geometry after contrastive and distillation training. Stacking learns *when* to trust each signal, improving AUROC from **0.670** (few-shot-only) to **0.746** (stacked) in repository exports.

6 Experimental setup

6.1 Cohort and label

MIMIC-IV v3.1 [6]; $n=2,000$; treatment feasibility label as in thesis; 64/16/20 split; seed 42.

6.2 Metrics

Primary: test AUROC; secondary: F1-macro, accuracy, MCC, AUPRC where exported. Bootstrap CIs from `exp_fewshot_best.json`.

6.3 Comparators

- Tuned tabular models (`exp_tabular_sota.csv`, `exp_final_best_model.csv`).
- Few-shot-only BEST and Stacked BEST (`exp_fewshot_best.json`).
- Exp1 RF and Proposed FewShot for controlled gap reporting (`exp1_results.csv`).

7 Results

Table 1–3 summarize repository exports.

The stacked model matches the best tabular export within **0.001** AUROC, while few-shot-only remains **0.077** below stacked, underscoring the value of classical fusion.

Table 1: Tuned tabular models (`exp_tabular_sota.csv`, unique configs).

Model	AUROC	F1 (macro)	Acc.	MCC
RandomForest	0.744	0.663	0.690	0.327
XGBoost	0.740	0.575	0.675	0.248
CatBoost	0.731	0.664	0.683	0.329
LightGBM	0.727	0.641	0.678	0.288

Table 2: Augmented best tabular row (`exp_final_best_model.csv`).

Model	AUROC	F1 (macro)	Acc.	MCC
aug-gaussian (best row)	0.747	0.673	0.700	0.348

8 Interpretability and deployment considerations

8.1 Tabular explanations

Random Forests admit Gini importance; SHAP TreeExplainer provides patient-level attributions on $\mathbf{x}$ [2]. These align with clinician expectations (e.g., creatinine, lactate).

8.2 Neural explanations

Attention weights over Symptom Tokens highlight which measurements drive representation; caution: attention is not causal. Few-shot prototypes can be visualized in 2-D projections for pedagogy.

8.3 Human-in-the-loop

We recommend threshold policies on $\hat{p}$, monitoring calibration drift, and subgroup reporting (age, sex, diagnosis node) before any real-world use [3].

9 Discussion

9.1 Representation trade-offs

Ablations (thesis Exp4) show intensity weighting matters and that bypassing tokens can increase AUROC in some configurations—honest reporting of such tensions strengthens scientific credibility.

9.2 Data scale

$n=2,000$ supports method comparison but limits transformer capacity; larger ICU corpora may change relative rankings [9].

9.3 Limitations

Single-center retrospective data; simplified outcome; stacking meta-learner fit on finite validation data; MCC for few-shot-only near zero reflects thresholding in export.

Table 3: Few-shot vs. stacked (`exp_fewshot_best.json`).

Strategy	AUROC	95% CI	F1 (macro)	Acc.	MCC
BEST_Stacked	0.746	[0.699, 0.797]	0.687	0.698	0.382
BEST_FewShot_GFA_CrossAttn	0.670	[0.612, 0.726]	0.386	0.628	0.000

Table 4: Exp1 controlled comparison (selected rows, `exp1_results.csv`).

Model	AUROC	F1 (macro)
RandomForest	0.762	0.679
Proposed_FewShot	0.611	0.572

9.4 Future work

Federated training of stacked models, multimodal notes + labs, and prospective validation.

10 Conclusion

Symptom Tokens and an ETHOS-style transformer provide a structured representation for early ICU labs and vitals, compatible with few-shot meta-learning. **Calibrated stacking** with a strong Random Forest closes most of the AUROC gap to the best tabular export (**0.746** vs. **0.747**) on a fixed MIMIC-IV benchmark, while naïve few-shot training without fusion remains much weaker in the Exp1 harness (**0.611** vs. **0.762**). These results argue for treating **representation design** and **fusion strategy** as first-class reporting dimensions in clinical deep learning.

Author contributions

M.S. designed tokens, models, and experiments; Dr. A. supervised.

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Competing interests

None declared.

Data and code availability

MIMIC-IV via PhysioNet; implementation in companion repository.

A Feature list (tabular branch)

Fifteen labs: creatinine, BUN, K^+ , Na^+ , Cl^- , HCO_3^- , hemoglobin, platelets, WBC, glucose, lactate, albumin, ALT, bilirubin, INR. Seven vitals: HR, SBP, DBP, MAP, temperature, SpO_2 , RR. Engineered: shock index, SOFA proxy, abnormal lab fraction.

B Hyperparameter pointers

See thesis `chapters/experiments.tex` and repository `config.yaml` for grids and search budgets.

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